CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Surface area

1. Find the total surface area of a solid hemisphere whose diameter is $2d$. (a) $3\pi d^2$ (b) $2\pi d^2$ (c) $\frac{1}{2}\pi d^2$ (d) $\frac{3}{4}\pi d^2$ CBSE 2026, 1 mark

SOLUTION

Radius $= \frac{2d}{2} = d$

Total surface area of a solid hemisphere $= 3\pi r^2 = 3\pi d^2$

Answer: (a)

2. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius r is $3\pi r^2$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2025, 1 mark

SOLUTION

A is true: the two flat circular faces fit together exactly and the curved surfaces make a complete sphere.

R is false: the surface area of a sphere is $4\pi r^2$. ($3\pi r^2$ is the total surface area of a solid hemisphere.)

Answer: (c)

3. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) $1:1$ (b) $1:4$ (c) $2:3$ (d) $3:2$ CBSE 2024, 1 mark

SOLUTION

Surface area of the sphere $= 4\pi r^2$

Each solid hemisphere has total surface area $3\pi r^2$, so the two together have $6\pi r^2$.

Ratio $= 4\pi r^2 : 6\pi r^2 = 2:3$

Answer: (c)

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4. A cylinder 5 cm high has a curved surface area of 94.2 cm². Find its radius, taking $\pi = 3.14$. (a) 2 cm (b) 3 cm (c) 2.9 cm (d) 6 cm CBSE 2023, 1 mark

SOLUTION

$2\pi rh = 94.2$: $2 \times 3.14 \times r \times 5 = 94.2$, so $31.4r = 94.2$ and $r = 3$ cm.

Answer: (b)

5. Find the curved surface area of a cone of height 24 cm and radius 7 cm. (a) 528 cm² (b) 1056 cm² (c) 550 cm² (d) 500 cm² CBSE 2023, 1 mark

SOLUTION

Slant height $l = \sqrt{24^2 + 7^2} = \sqrt{625} = 25$ cm

Curved surface area $= \pi rl = \frac{22}{7} \times 7 \times 25 = 550$ cm²

Answer: (c)

6. A conical cavity of height 24 cm is hollowed out of a solid cylinder of height 30 cm and radius 7 cm. The cone has the same radius as the cylinder. Find the total surface area of the solid that is left. CBSE 2022 Term 2, 4 marks

SOLUTION

Slant height of the cone $l = \sqrt{24^2 + 7^2} = 25$ cm

The remaining solid has three surfaces: the curved surface of the cylinder, one flat circular end, and the inner curved surface of the cone.

Total surface area $= 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l) = \frac{22}{7} \times 7 \times (60 + 7 + 25) = 22 \times 92$

Answer: 2024 cm²

7. A wooden article is made by scooping a hemisphere out of each end of a solid cylinder. The cylinder is 10 cm high and its base has radius 3.5 cm. Find the total surface area of the article. Take $\pi = \frac{22}{7}$. CBSE 2018, 3 marks

SOLUTION

The surface consists of the curved surface of the cylinder and the curved surfaces of the two hemispherical hollows.

Total surface area $= 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$

$= 2 \times \frac{22}{7} \times 3.5 \times (10 + 7) = 22 \times 17$

Answer: 374 cm²

8. A tent is a cylinder with a cone of the same diameter on top. The cylinder is 2.1 m high and 3 m in diameter, and the slant height of the cone is 2.8 m. Canvas costs ₹500 per square metre. Find the cost of the canvas for the tent. Take $\pi = \frac{22}{7}$. CBSE 2016, 3 marks

SOLUTION

Radius $r = 1.5$ m, $h = 2.1$ m, $l = 2.8$ m. Canvas covers the curved surface of the cylinder and of the cone.

Area $= 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 1.5 \times (4.2 + 2.8) = \frac{22}{7} \times 1.5 \times 7 = 33$ m²

Cost $= 33 \times 500$

Answer: ₹16,500

9. After a flood, 50 schools agree to share equally the cost of canvas for 1500 tents. Each tent is a cylinder of base radius 2.8 m and height 3.5 m, with a cone of the same radius and height 2.1 m on top. Canvas costs ₹120 per square metre. Find the amount each school pays. Take $\pi = \frac{22}{7}$. CBSE 2016, 4 marks

SOLUTION

Slant height of the cone $l = \sqrt{2.8^2 + 2.1^2} = \sqrt{7.84 + 4.41} = \sqrt{12.25} = 3.5$ m

Canvas for one tent $= 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 2.8 \times (7 + 3.5) = 8.8 \times 10.5 = 92.4$ m²

Cost of one tent $= 92.4 \times 120 = 11088$ rupees. Cost of 1500 tents $= 11088 \times 1500 = 16632000$ rupees.

Share of each school $= \frac{16632000}{50}$

Answer: ₹3,32,640

2 Volume

10. A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take $\pi = \frac{22}{7}$. CBSE 2026, 3 marks

SOLUTION

Radius $r = \frac{7}{2}$ cm. Height of the cylinder $= 20 - 2r = 20 - 7 = 13$ cm.

Volume of the cylinder $= \pi r^2 h = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 13 = 500.5$ cm³

The two hemispheres make one sphere: $\frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \frac{539}{3} \approx 179.67$ cm³

Total volume $= 500.5 + 179.67$

Answer: about 680.17 cm³

11. A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the height of the cylindrical part. Take $\pi = \frac{22}{7}$. CBSE 2025, 3 marks

SOLUTION

Let the radius be r m. Then the height of the cylinder is $2r$ m.

Volume $= \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3$

$\frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{1408}{21}$, so $\frac{176}{21}r^3 = \frac{1408}{21}$

$r^3 = 8$, so $r = 2$ and the height of the cylinder is $2r = 4$.

Answer: 4 m

12. The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a) $\frac{4\pi}{3}$ cu cm (b) $\frac{5\pi}{3}$ cu cm (c) $\frac{8\pi}{3}$ cu cm (d) $\frac{2\pi}{3}$ cu cm CBSE 2024, 1 mark

SOLUTION

The largest cone has base diameter 2 cm and height 2 cm, so $r = 1$ cm and $h = 2$ cm.

Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 1 \times 2 = \frac{2\pi}{3}$ cu cm

Answer: (d)

13. A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm³. Find the two radii. CBSE 2024, 3 marks

SOLUTION

Let the outer and inner radii be R and r. Then $R - r = 1$.

Volume of metal $= \pi(R^2 - r^2)h$: $\frac{22}{7} \times (R + r)(R - r) \times 14 = 176$

$44(R + r) = 176$, so $R + r = 4$.

Solving with $R - r = 1$: $R = 2.5$ and $r = 1.5$.

Answer: outer radius 2.5 cm, inner radius 1.5 cm

14. A room is a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds $\frac{1408}{21}$ m³ of air. Find the total height of the room. Take $\pi = \frac{22}{7}$. CBSE 2023, 3 marks

SOLUTION

Let the radius be r m. Then the cylinder is $2r$ m high.

Volume $= \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3 = \frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{176}{21}r^3$

$\frac{176}{21}r^3 = \frac{1408}{21}$, so $r^3 = 8$ and $r = 2$.

Total height = cylinder + dome $= 2r + r = 3r$

Answer: 6 m

15. An ice-cream cone has radius 3 cm and height 12 cm. The lowest part of the cone, one-sixth of its volume, is left empty. The rest is filled with ice-cream, with a hemisphere of ice-cream on top. Find the volume of the ice-cream. Take $\pi = 3.14$. CBSE 2023, 3 marks

SOLUTION

Volume of the cone $= \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 9 \times 12 = 36\pi$ cm³

Ice-cream inside the cone $= \frac{5}{6} \times 36\pi = 30\pi$ cm³

Hemisphere on top $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \pi \times 27 = 18\pi$ cm³

Total $= 48\pi = 48 \times 3.14$

Answer: 150.72 cm³

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